Panchip Microelectronics Co., Ltd.

PAN3029/3060 Automatic SF Identification Application Guide

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Document Description

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Revision History

Version	Revision Date	Description
V1.0	2023.06	Initial version
V1.1	2024.03	Added instructions for using Smart Search
V1.2	2025.06	Updated the automatic SF recognition process

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1 Function Introduction

The PAN3029/3060's intelligent automatic spreading factor (Auto-SF) function is a core feature designed specifically for lightweight ChirpIOT gateways. By dynamically analyzing the SF parameters (7-12) in the channel in real time, it enables a single receiver to accommodate multi-rate signals, resolving the resource bottleneck of "multi-channel parallel reception" in traditional solutions.

2 Principle of automatic SF recognition

Receiver Automatic Spreading Factor (SF) Identification Principle

1. Spreading Factor (SF) Definition:

The device supports spreading factors from SF5 to SF12.

SF defines the FFT length used by the receiver (RX) to demodulate linear frequency modulation (chirp) signals. Specifically, the relationship is: FFT length = 2^SF.

Thus:

When SF = 5, FFT length = 2^5 = 32 samples per chirp.

When SF = 6, FFT length = 2^6 = 64 samples per chirp.

...And so on.

When SF = 12, FFT length = 2^12 = 4096 samples per chirp.

2. Automatic SF Identification Mechanism:

The receiver can be configured to perform a cyclic search within a specified set of SFs (for example, searching only SF5 and SF6).

On each target SF, the RX dwells for a duration equal to the duration of two complete chirps corresponding to that SF (i.e., 2 * 2^SF samples).

Search Process Example (SF5 & SF6):

The receiver dwells on SF5 for 2 * 2^5 = 64 samples (equivalent to two SF5 chirps).

If no valid chirp signal is detected within this window, the RX switches to SF6.

The receiver dwells on SF6 for 2 * 2^6 = 128 samples (equivalent to two SF6 chirps).

If no signal is detected, the RX switches back to SF5 and repeats the cycle.

3. Transmitter Preamble Length Requirements:

To ensure the RX can successfully acquire the TX signal during the search cycle, the preamble sent by the TX must be long enough to cover the time the receiver dwells on non-target SFs.

Calculating the minimum preamble length (number of samples):

Calculating the total search cycle duration (T_cycle): Add the dwell times (2 * 2^SF) for all configured search SFs.

Example (SF5, SF6): T_cycle = (2 * 2^5) + (2 * 2^6) = 64 + 128 = 192 samples.

Example (SF5, SF6, SF7): T_cycle = (2 * 2^5) + (2 * 2^6) + (2 * 2^7) = 64 + 128 + 256 = 448 samples.

Example

Number of chirps converted to target SF (N_min_chirps): N_min_chirps = T_cycle / (2^Target_SF)

Example (target SF=5): N_min_chirps = 192 / 32 = 6 chirps

Example (target SF=6): N_min_chirps = 192 / 64 = 3 chirps (for SF5 & SF6 cycles)

Example (target SF=5, searching SF5/6/7): N_min_chirps = 448 / 32 = 14 chirps

Example (target SF=7, search SF5/6/7): N_min_chirps = 448 / 128 = 3.5 Chirps (actual value needs to be rounded up)

Adding Margin: To account for timing variations and ensure robustness, add additional chirps to N_min_chirps as margin (for example, adding 2 chirps is a common practice).

Example (target SF=5, search SF5/6, +2 chirps): N_preamble = 6 + 2 = 8 chirps

Example (target SF=5, search SF5/6/7, +2 chirps): N_preamble = 14 + 2 = 16 chirps

Example (target SF=7, search SF5/6/7, +2 chirps): N_preamble = ceil(3.5) + 2 = 4 + 2 = 6 chirps^ (ceil(3.5) = 4)

4. Key Conclusion:

The core steps for determining the number of chirps required for the TX preamble are:

Based on the configured search SF list, calculate the total search cycle duration, T_cycle = Σ (2 * 2^SF) (summed over all search SFs).

Calculate the minimum number of chirps under the target SF, N_min_chirps = T_cycle / (2^Target_SF).

Add the engineering margin, Margin, to N_min_chirps to obtain the final required number of preamble chirps: N_preamble = N_min_chirps + Margin.

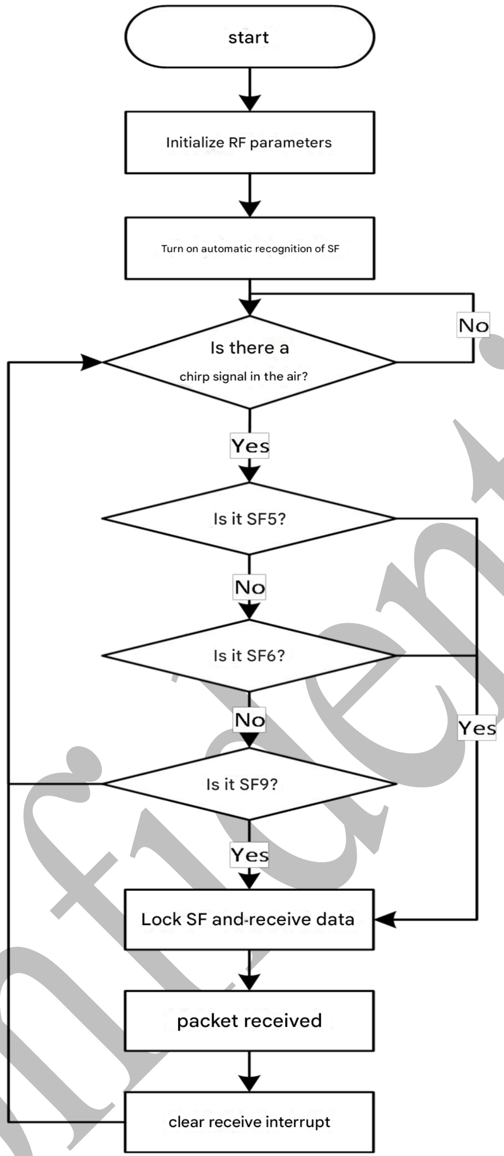
5. Receiver Automatic SF Identification Flowchart

As shown below, after the PAN3029/3060 has the automatic SF identification function enabled and enters receive mode, if it detects a preamble signal in the air, the receiver will sequentially determine the SF according to the automatic SF identification configuration. The configuration registers are as follows:

SF Search Enable Configuration:

- [Page1][0x2D][BIT0]: 1 to enable SF5 identification; 0 to disable SF5 identification;
- [Page1][0x2D][BIT1]: 1 to enable SF6 identification; 0 to disable SF6 identification;
- [Page1][0x2D][BIT2]: 1 to enable SF7 identification; 0 to disable SF7 identification;
- [Page1][0x2D][BIT3]: 1 to enable SF8 identification; 0 to disable SF8 identification;
- [Page1][0x2D][BIT4]: 1 to enable SF9 identification; 0 to disable SF9 identification;
- [Page1][0x2D][BIT5]: 1 to enable SF10 identification; 0 to disable SF10 identification;
- [Page1][0x2D][BIT6]: 1 to enable SF11 identification; 0 to disable SF11 identification;

- [Page1][0x2D][BIT7]: 1 to enable SF12 identification; 0 to disable SF12 identification.



In the above flowchart, the SF5/SF6/SF9 automatic identification function is enabled, and the corresponding [Page1][0x2D] register is configured to 0x13.

3 Software Design Reference

When using this function, you need to configure it on both the transmitter and receiver.

SDK interface function:

```
/**
  * @brief Enables the corresponding SF search based on the SF set.
  * @param pSfPream: Pointer to an array of AutoSfPream_t structures.
  * @param SF_Num: Number of SFs to search.
  * @note: The auto-search function cannot be used simultaneously with the CAD function!
  */
static void RF_EnableAutoSF(AutoSfPream_t *pSfPream, int SF_Num);
/**
  * @brief Disables the SF receive adaptive function.
  */
void RF_DisableAutoSF(void);
/**
  * @brief Calculates the number of FFT points based on a given SF value
  * @param sf : The passed SF value
  * @return : The number of FFT points required to sample a single chirp symbol
  * @note This function calculates the number of FFT points based on the SF value.
  The SF value range is 5 to 12.*/
uint16_t GetFFTPointsBySF(RfSpreadFactor_t sf);

typedef struct
{
        RfSpreadFactor_t Sf;            //!< Spreading factor
        uint16_t TxPreamLen;            //!< Tx preamble length according to SF
}AutoSfPream_t;

/**
  * @brief Calculates the TxPreamLen for each SF in a given SF set.
  * @param pSfPream: Pointer to an array of AutoSfPream_t structures.
  * @param SF_Num: Number of SFs in the set.
```

```
 * @note This function calculates the TxPreamLen value for each SF based on the given SF set.
 */
void CalculateTxPreamLenBySF(AutoSfPream_t *pSfPream, int SF_Num);
/**
 * @brief Gets the SF configuration for a received packet.
 * @return Returns the SF value for the current received packet.
 * @note The SF value is in the upper 4 bits of register [Page1][0x7C].
 */
uint8_t RF_GetPktSF(void);
```

3.1 Receiver Configuration

Before receiving, enable automatic SF identification by calling the RF_EnableAutoSF() function. The PAN3029/3060 can then receive packets with different SF signals on the same channel. After receiving a packet, the user can retrieve the SF configuration parameters for the packet using the RF_GetPktSF() function.

To disable this feature, configure the receiver to call the RF_DisableAutoSF function. This allows the chip to only receive data with the specified SF signal on the same channel, and then continue receiving data as normal.

3.2 Transmitter Configuration

Before transmitting, the transmitter needs to configure different preamble lengths based on different SF values. The CalculateTxPreamLenBySF(AutoSfPream_t *pSfPream, int SF_Num) function can be used to calculate the corresponding preamble lengths for different SFs.

When automatic SF identification is disabled, the transmitter's preamble length is typically set to 8. When automatic SF identification is enabled, the transmitter's preamble length is typically increased, which increases the chip's over-the-air transmission time. Generally, the more SFs that require automatic identification, the longer the preamble length the transmitter needs to send.

4 Notes

Using the Automatic SF Recognition function, you must set the number of symbols detected by CAD to RF_CAD_01_SYMBOL. Normally, the Automatic SF Recognition function is not used together with the CAD function.